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Sexual reproduction of *Acropora* reef corals at Moorea, French Polynesia

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Abstract Little information is available on reproductive processes among corals in isolated central Pacific reef regions, including French Polynesia. This study examined the timing and mode of sexual reproduction for *Acropora* reef corals at Moorea. Spawning was observed and/or inferred in 110 *Acropora* colonies, representing 12 species, following full moon periods in September through November 2002. Gamete release was observed and inferred in four species of *Acropora* between 9 and 13 nights after the full moon (nAFM) in September 2002. Twelve *Acropora* spp. spawned gametes between 5 and 10 nAFM in October 2002, with six species spawning 7 nAFM and four species spawning 9 nAFM. In November 2002, spawning of egg and sperm bundles was observed and inferred in 27 colonies of *Acropora austera*, 6 nAFM. These are the first detailed records of spawning by *Acropora* corals in French Polynesia.

Keywords Spawning · *Acropora* reef corals · Moorea · French Polynesia

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Introduction

Following the discovery of multispecific broadcast spawning of scleractinian reef corals on the Great Barrier Reef (GBR) (Harrison et al. 1984), there has been a substantial increase in research on coral reproduction patterns in many reef regions (reviewed by Harrison and Wallace 1990; Richmond and Hunter 1990; Shlesinger et al. 1998; Wilson and Harrison 2003). Of the hermaphroditic scleractinian corals that have been studied to date, the great majority are hermaphroditic and broadcast eggs and sperm into the water column for external fertilisation and development, show periodicity in their gamete release and have relatively brief annual breeding seasons (Harrison and Wallace 1990; Richmond and Hunter 1990; Shlesinger et al. 1998). However, the timing and degree of spawning synchrony within and among coral species can vary widely in different locations and along latitudinal gradients (e.g. Shlesinger and Loya 1985; Oliver et al. 1988; Baird et al. 2002; Wilson and Harrison 2003; Mangubhai and Harrison 2005).

On the GBR, over 140 broadcast spawning reef coral species are known to participate in annual multispecific, synchronous ‘mass spawning’ events, during the week following the full moon periods in late spring and early summer (Harrison et al. 1984; Willis et al. 1985; Babcock et al. 1986; Oliver et al. 1988; Harrison 1993). Mass spawning has also been reported in the Dampier Archipelago (Simpson 1985) and Houtman Abrolhos Islands (Babcock et al. 1994) in western Australia, and on reefs at Akajima Island in southern Japan (Hayashibara et al. 1993). More recently, multispecific synchronous coral spawning has been observed in the Solomon Islands (Baird et al. 2002) and in Singapore (Guest et al. 2002, 2005), revealing that a high degree of spawning synchrony may also be characteristic of some equatorial coral assemblages, in the absence of major fluctuations in environmental variables. Some degree of multispecific spawning has been reported in the Caribbean where coral spawning generally occurs from July to September

each year (Szmant 1986; Richmond and Hunter 1990; Soong 1991; Gittings et al. 1992; van Veghel 1993, 1994; Steiner 1995). Extended reproductive seasons have been recorded at Eilat in the northern Red Sea, Hawaii, the Gulf of Mexico and in some central and eastern Pacific reef regions (Shlesinger and Loya 1985; Harrison and Wallace 1990; Richmond and Hunter 1990; Glynn et al. 1994; Shlesinger et al. 1998). A greatly extended reproductive period has more recently been documented for equatorial coral communities in Kenya (Mangubhai and Harrison 2005).

In contrast, very little information is available on sexual reproductive patterns among corals in many isolated central Pacific reef areas, including French Polynesia. To obtain a global perspective of the reproductive and life-history patterns of scleractinian corals, sexual reproductive processes must be studied in areas that have received little or no investigation. French Polynesia represents an appropriate location to study coral reproduction because (1) the coral reefs of French Polynesia are insular and oceanic, and therefore relatively isolated (2) French Polynesia represents the eastern boundary of moderate scleractinian reef coral biodiversity (Veron 1995), and (3) there are no detailed records of coral spawning in this region. This paper presents data on the timing and mode of sexual reproduction for 12 *Acropora* species in 2002 at Moorea, which represent the first detailed coral spawning records for French Polynesia.

Methods

Study area

Moorea (17°30'S 149°50'W), part of the Society Archipelago (Windward Island group), is a high volcanic island that lies 25 km to the northwest of Tahiti. Tides are

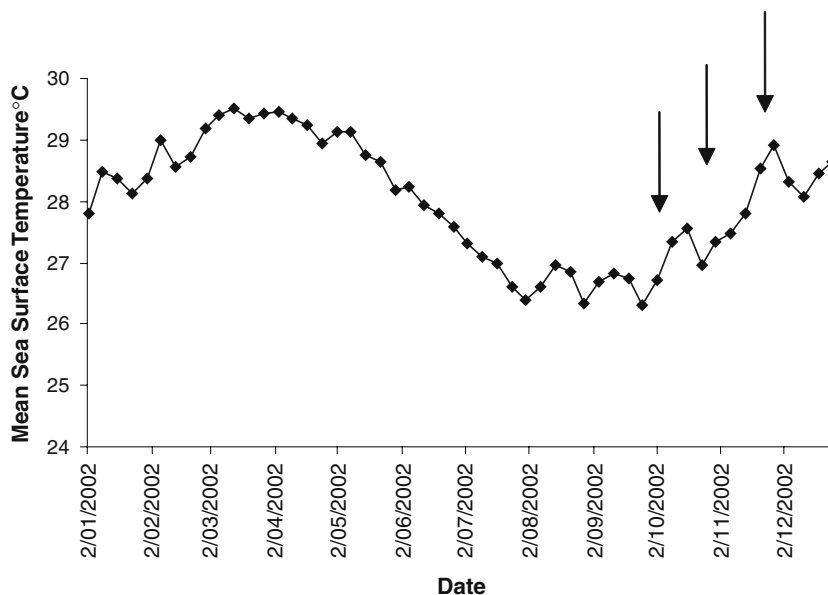
semi-diurnal with a tidal range that rarely exceeds 40 cm (Adjeroud 1997). Four study sites were established in the north and northwest part of Moorea at Vaipahu reef lagoon (17°28.945'S 149°50.980'W) and outer reef slope (17°28.795'S 149°51.135'W) and at Tiahura reef lagoon (17°29.155'S 149°54.019'W) and outer reef slope (17°28.987'S 149°54.068'W). The depth range was 1–3 m at reef lagoon sites and 6–20 m on the outer reef slope. Sea temperature ranges from a minimum of 26.3°C in winter to a maximum of 29.5°C in summer (Fig. 1).

Timing of reproduction

A total of 94 *Acropora* coral colonies, representing seven species were initially tagged and sampled in May and June 2002, in order to determine their reproductive status. These colonies were subsequently re-sampled approximately monthly during September through November 2002, and during 2003, to determine their gametogenic cycles. At each sample period, small samples (approximately 4–8 cm long) were taken from each colony and fixed in 10% formalin in seawater, then decalcified and dissected. Preliminary analysis of these samples has revealed one gametogenic cycle per year.

To determine the timing and synchrony of spawning, an additional 125 *Acropora* coral colonies representing 12 species were tagged and repeatedly sampled or collected for spawning in aquaria, during September through November 2002. All 125 colonies were gravid when first sampled. Oocytes usually become pigmented close to the time of spawning and are readily visible in broken sections of coral (Harrison et al. 1984). In colonies where mature eggs were detected, samples were taken daily or every few days in situ, until spawning on the reef was inferred by the disappearance of mature gametes between sequential samples.

Fig. 1 Weekly mean sea surface temperature readings for Tahiti 2002. Arrows indicate the main spawning periods for *Acropora* species in September, October and November. Source: IGOSS-nmc sea temperature data, courtesy of the Lamont-Doherty Climate Center at Columbia University (<http://rainbow.ldeo.columbia.edu/>)



From each study site, portions of reproductively mature coral colonies (including portions of some tagged field colonies) were collected every few weeks from September through November 2002, and maintained in laboratory aquaria to determine the timing and behaviour of gamete release. In order to confirm that spawning observations in aquaria were indicative of coral spawning patterns occurring on reefs, within 6–48 h after spawning was observed in aquaria, tagged colonies of these species at the reef study sites were re-sampled.

Results and discussion

A total of 110 of the 125 coral colonies representing 12 *Acropora* spp. were recorded to release gametes nocturnally following the full moon periods in September, October and November 2002 (Table 1, Fig. 2). All 12 *Acropora* species were simultaneous hermaphrodites and

spawned egg–sperm bundles. During September 2002, gamete release was observed or inferred in four species between 9 and 13 nights after the full moon (nAFM). Nine species were observed spawning in aquaria between 5 and 10 nAFM in October, with minor spawning recorded on the night of the full moon (Table 1). On 26th November 2002 (6 nAFM), spawning of egg and sperm bundles was observed in portions of 11 colonies of *Acropora austra* in aquaria. Spawning on the sixth nAFM in five of these 11 colonies and in 11 additional colonies of this species was confirmed in the field from the disappearance of mature gametes between sequential samples taken on the 26th and 27th November 2002.

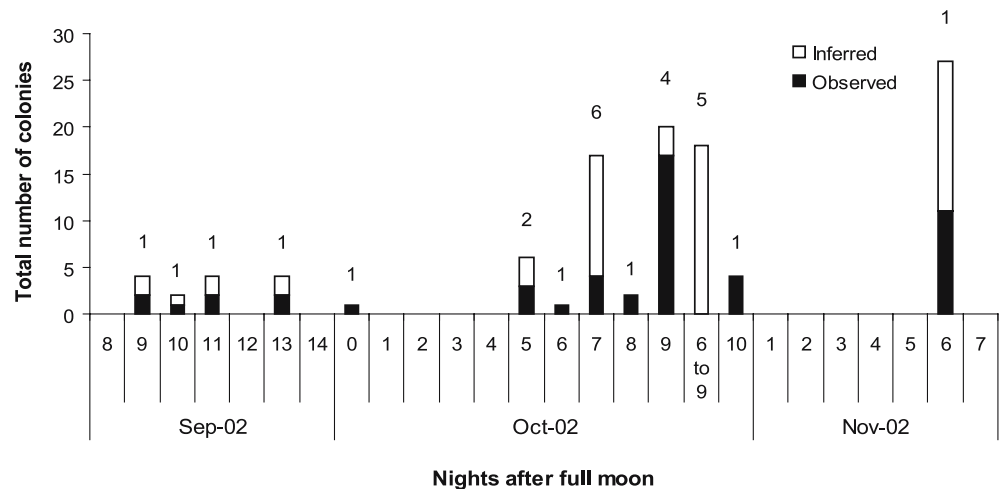
Although most gravid *Acropora* colonies were observed or inferred to spawn, the spawning period for 15 colonies, representing six *Acropora* species, could not be determined (Table 1). These colonies had large oocytes or coloured eggs when they were last sampled around the new or full moon period in November 2002, but

Table 1 *Acropora* coral species recorded spawning after full moon periods in September to November in 2002

Species	Total number of gravid colonies	Total number of colonies observed spawning	Total number of colonies inferred spawning	Percent spawning	Spawning periods (number of colonies spawning)		
					September	October	November
<i>Acropora acuminata</i>	2	1	1	100	10(2)		
<i>A. austra</i>	32	11	16	84			6(27)
<i>A. cytherea</i>	8	2	2	50		7(3), 6–9(1)	
<i>A. globiceps</i>	3	3	0	100		9(3)	
<i>A. hyacinthus</i>	14	2	9	79		7(6), 6–9(5)	
<i>A. lutkeni</i>	28	10	17	96		7(4), 9(13), 6–9(10)	
<i>A. polystoma</i>	1	1	0	100		9(1)	
<i>A. pulchra</i>	4	2	2	100	9(4)		
<i>A. retusa</i>	12	7	4	92	11(4)	6(1), 7(1), 6–9(1), 10(4)	
<i>A. secale</i>	11	6	5	100	13(4)	5(4), 7(1), 8(2)	
<i>A. striata</i>	7	2	4	86		0(1), 5(2), 7(3)	
<i>A. cf. divaricata</i>	3	2	1	100		9(2), 6–9(1)	

Bold numbers indicate nights after full moon when spawning was observed/inferred; numbers in () refer to number of colonies spawning within each spawning period

Fig. 2 Summary of spawning results for *Acropora* in 2002. Numbers above each column refer to the number of species spawning



could not be re-sampled. These colonies may have spawned after the full moon period in November or December 2002.

These are the first detailed records of spawning by *Acropora* corals in French Polynesia. The majority of *Acropora* species studied spawn during a single lunar period, between 5 and 10 nAFM in October. *Acropora acuminata* and *A. pulchra* spawned in September, *A. austra* in November, and *A. retusa* and *A. secale* spawned following the full moon periods in both September and October. These spawning patterns are similar to those recorded on the GBR where split spawning over a 2-month period has been observed in some species, in some years (Willis et al. 1985). Some degree of spawning synchrony was observed and inferred for some colonies of *Acropora* (six species spawned 7 nAFM and four species spawned 9 nAFM in October), and 27 colonies of *A. austra* spawned synchronously 6 nAFM in November (Table 1).

The synchrony, predictability and brevity of coral spawning on the GBR appear to be linked to successive environmental cues which operate on increasingly finer time scales (Babcock et al. 1986). Babcock et al. (1986) suggested that the timing of mass spawning on the GBR may ultimately be due to the advantage of increased fertilisation during periods of reduced tidal currents and low water volume associated with neap tides. However, in contrast to the large tidal ranges on the GBR, tidal range in Moorea rarely exceeds 40 cm, hence tidal fluctuation may not exert a strong influence on the timing of coral spawning in Moorea. Spawning in Moorea occurred during a period of rising sea temperatures (Fig. 1), in the months between dominant trade wind shifts, and prior to the onset of the wet season. Hence, the timing of sexual reproduction for *Acropora* reef corals in Moorea may be influenced by rising sea temperatures or increased solar insolation during these seasonal changes. Further research is required in order to determine whether the spawning patterns observed in 2002 are typical of spawning patterns in other years, and to investigate the relationship between the timing of spawning and potential proximate cues.

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